SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B Tech Integrated Stream: Computer Science | Year: II Semester: IV

Subject: Chemistry-II | Time: 3 hrs (2:00 PM to 5:00 PM)

Date: 25 / 04 / 23 | No. of Pages:3

Marks: 100

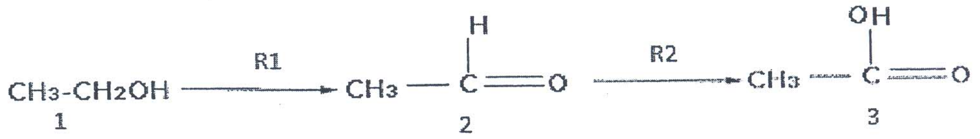
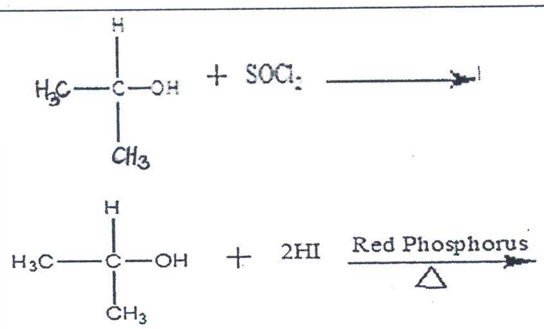
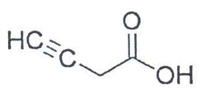
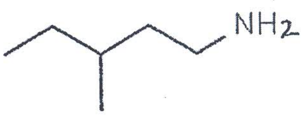
Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**
8) **Atomic weights:** H=1, C=12, O=16, Al=27, N=14, Li = 6.9, Na=23, Mg=24, P=31, S=32, Cl=35.5, Ca=40, Ni=58.7, Fe=55.85, K=39, B=10.8, Cu= 63.54, Mg=24, Cr=52

Q1		Answer briefly:	[20]
CO-1; SO-1 BL-2	a.	Define the term electron affinity (EA). Why is the first EA a positive energy value and the second EA a negative energy value.	[5]
CO-2 ; SO-1 BL-1	b.	Explain the concept of a mole w.r.t Avogadro's Number. Calculate number of oxygen atoms in a sample of 2.85 mol $CaCO_3$.	[5]
CO-3; SO-1 BL-3	c.	Illustrate the structure of the following compounds: i. 3-methyl butan-2-one ii. 2,5-dimethyl-1,5-hexadiene iii. m-chloro benzaldehyde iv. 3-methylbutanol v. 1-bromo-1,2-dichloropropane	[5]
CO-4; SO-1 BL-1	d.	Write a short note on multiple covalent bonds	[5]
Q2 CO-4; SO-1 BL-1	a.	Explain carboxylic acid functional group with respect to identification, 2 methods of preparation and 2 reactions.	[5]
CO-2; SO-1 BL-3	b.	i) What do you understand by dilution? Are the moles of substance before and after dilution same? Show the dilution formula.	[5]

		ii) What volume of 5 M NaOH is needed to create a 100 mL solution of 1 M NaOH?	
CO-1; SO-1 BL-1	c.	Define effective nuclear charge. Estimate Z_{eff} of outermost s electron and 3d electron for Cr (Z=24).	[5]
CO-2; SO-1 BL-3,5	d.	i) Calculate the weight of a sample of copper sulphate pentahydrate ($CuSO_4.5H_2O$) that contains 15 x 10^{18} molecules of it. ii) Which has more atoms? Explain, 1mol of carbon (C) or 1mol of Magnesium (Mg)	[5]
Q3 CO-3; SO-1 BL-1	a.	Explain covalent bond and its types with an appropriate example(s).	[5]
CO-4; SO-1 BL-3	b.	How many total number of atoms are there in 2.00 moles of : (i) CO_2 (ii) SO_2	[5]
CO-3; SO-1 BL-1	c.	Define octet rule. Illustrate Lewis structure of any 2 molecules that violates octet rule.	[5]
CO-4; SO-1 BL-3	d.	A solution was prepared by dissolving 40 g KCl in water such that total volume of solution is 500 ml. Density of solution is 1.13 g/ml. Calculate molality of solution.	[5]
Q4 CO-4; SO-1 BL-1	a.	Discuss aldehyde & ketone functional group with respect to 1 identification test, 2 method of preparation and 2 reaction of each.	[5]
CO-3; SO-1 BL-1,2	b.	Define functional group and give examples with structures of any 4 functional groups	[5]
CO-3; SO-1 BL-4	c.	How many grams of potassium are there in 5.4g of $K_2Cr_2O_7$?	[5]
CO-3; SO-1 BL-2	d.	Write any two reactions of aldehydes and ketones with examples.	[5]
Q5 CO-1; SO-1 BL-4	a.	Define the term atomic radii and discuss its trend across a period. Predict the period, group and block of elements having atomic number 15 and 19.	[5]
CO-4; SO-1 BL-1,4	b.	What do understand by gram equivalents. Calculate gram equivalents of 5.0g H_3PO_4 and 3.9g of $Mg(OH)_2$.	[5]
CO-1; SO-1 BL-1,4	c.	What do you understand by iso-electronic species? Arrange O^{2-} ,Na^+, F^- and Mg^{2+} in decreasing order of their atomic radius and justify your answer	[5]
CO-4; SO-1 BL-3	d.	Discuss the classification of solutions w.r.t different solvents used for making the solution and give appropriate examples	[5]
Q6 CO-3; SO-1 BL-1,4	a.	"Co-ordinate bond is the special case of covalent bond" justify. Explain formation of NH_4^+.	[5]

CO-4; SO-1 BL-4	b.	A solution was prepared by dissolving 42 g Na_2SO_4 in water such that total volume of solution is 220 ml. Density of solution is 0.891 g/ml. Calculate mole fraction of both the components of solution.	[5]
CO-3; SO-1 BL-2	c.	 In the above reaction identify: i) functional groups of 1, 2 and 3 ii) Names of 1,2 and 3 iii) type of reaction R1 and R2	[5]
CO-4; SO-1 BL-1	d.	Define ppm and ppb. A solution was prepared by dissolving 0.4 moles $CaSO_4$ in water such that total volume of solution is 500 ml. Find concentration in ppm and ppb.	[5]
Q7 CO-3; SO-1 BL-2	a.	 In the above reactions: i) Draw structures of the product formed ii) Write the names of the reactants and products	[5]
CO-3; SO-1 BL-2	b.	Explain carboxylic acid functional group with respect to identification, 2 methods of preparation and 2 reactions.	[5]
CO-3; SO-1 BL-2	c.	Identify functional groups and write IUPAC names of following compounds: i)  ii) 	[5]
CO-2; SO-1 BL-3	d.	Show Lewis structure of BF_3 and CH_4	[5]